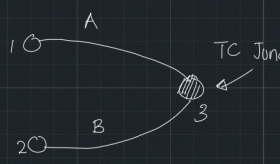
10/2

Thermocouple



$$V_3 - V_1 = \int_{T_1}^{T_3} S_A \, dT \quad (I)$$

$$V_2 - V_3 = \int_{T_3}^{T_2} S_B \, dT \quad (II)$$

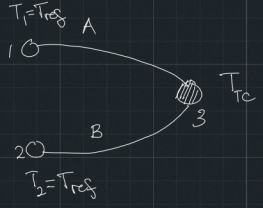
$$(I)+(II): \; V_2 - V_1 = \int_{T_1}^{T_3} S_A \, dT + \int_{T_3}^{T_2} S_B \, dT = -\int_{T_2}^{T_3} S_B \, dT$$

$$V_2 - V_1 = \int_{T_1}^{T_3} S_A \, dT - \int_{T_2}^{T_3} S_B \, dT = \int_{T_1}^{T_3} (S_A - S_B) \, dT \quad (\text{if } T_1 = T_2)$$

$$V_2 - V_1 = (\bar{S}_A - \bar{S}_B)(T_3 - T_1) \quad \left(\text{if } S_A = \bar{S}_A \text{ and } S_B = \bar{S}_B\right)$$

$$T_3 = T_1 + \frac{V_2 - V_1}{\bar{S}_A - \bar{S}_B}$$

Standard TC tables



The TC tables will give:

func.: $V_2 - V_1$

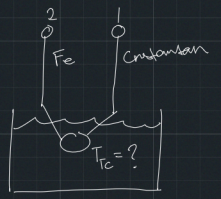
input: T_{TC}

for

$T_1 = T_2 = T_{ref}$ (usually 0°C)

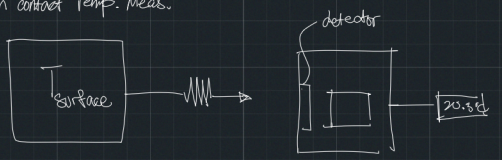
Type	Wires	Temp Range	Typical Application
B	Pt, 30% Rh(+) vs Pt, 6% Rh(−)	0 → 1820	Highest Temp. Non oxidizing env. High Temp
E			
J			
K			
R			
S			
T			

Type J TC



$$V_2 - V_1 = +5.850 \text{ mV}$$

Non contact Temp. Meas.



$$T = \left(\frac{\epsilon}{\epsilon_0}\right)^{\frac{1}{4}}$$

Set emissivity setting of device

Manufacturers have rough values for common surfaces

Mechanical Temp Meas. Devices

Exploits the fact that the density of a material depends on the temperature